

Homework 12: Soil, Impressions & Advanced Prints

Paper track. Pairs with Home Lab 12. A soil texture triangle helps.

Part A: Soil texture triangle (2 pts each)

Name the texture class:

1. 70% sand, 20% silt, 10% clay
2. 15% sand, 65% silt, 20% clay
3. 33% sand, 33% silt, 34% clay
4. 45% sand, 40% silt, 15% clay
5. 10% sand, 30% silt, 60% clay

Part B: Soil facts (1 pt each)

1. Particle-size range for sand, silt, and clay.
2. What causes soil to be red? Dark brown/black? Gray?
3. Is soil class evidence or individual evidence?
4. A soil is 100% minus 55% sand minus 30% silt — what's the clay %? Then name the class.

Part C: Fingerprints (2 pts each)

1. List the eight IAFIS patterns.
2. Delta count for arch, loop, whorl.
3. A print shows two loops swirled together like a yin-yang. Which pattern?
4. Latent, patent, plastic — define each, and give one way to develop a latent.
5. Define: dermatoglyphics, dactyloscopy, adermatoglyphia, hexadactyly.
6. What does each stand for: AFIS, IAFIS, CODIS, NCIC? Which one is a DNA database?

Part D: Shoeprint (3 pts)

A crime-scene shoeprint and a suspect's shoe both show a men's size 10, a herringbone tread, and heavy wear on the outer heel. The crime-scene print also shows a **triangular gouge across the third tread bar**, which is present on the suspect's actual sole.

1. Which features are class characteristics? Which is an individual characteristic?
 2. Does the size + tread + wear match prove it's the same shoe? Does the gouge?
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